In [7]:

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report


X = pd.read_csv('X_final.csv')
y = pd.read_csv('y_final.csv')

y = y['AQI_Bucket']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)

y_pred_knn = knn.predict(X_test)

print("\nKNN RESULTS")
print("Accuracy:", accuracy_score(y_test, y_pred_knn))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_knn))
print("Classification Report:\n", classification_report(y_test, y_pred_knn))
lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)

print("\nLOGISTIC REGRESSION RESULTS")
print("Accuracy:", accuracy_score(y_test, y_pred_lr))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_lr))
print("Classification Report:\n", classification_report(y_test, y_pred_lr))

plt.figure(figsize=(10, 8))
sns.heatmap(X.corr(), annot=True, cmap='coolwarm')
plt.title("Feature Correlation Heatmap")
plt.show()



plt.figure(figsize=(8, 5))
sns.countplot(x=y)
plt.title("AQI Category Distribution")
plt.xticks(rotation=45)
plt.show()



cm_knn = confusion_matrix(y_test, y_pred_knn)

plt.figure(figsize=(6, 5))
sns.heatmap(cm_knn, annot=True, fmt='d', cmap='Blues')
plt.title("KNN Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()


cm_lr = confusion_matrix(y_test, y_pred_lr)

plt.figure(figsize=(6, 5))
sns.heatmap(cm_lr, annot=True, fmt='d', cmap='Greens')
plt.title("Logistic Regression Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

knn_acc = accuracy_score(y_test, y_pred_knn)
lr_acc = accuracy_score(y_test, y_pred_lr)

models = ['KNN', 'Logistic Regression']
accuracies = [knn_acc, lr_acc]

plt.figure(figsize=(6, 4))
plt.bar(models, accuracies)
plt.title("Model Accuracy Comparison")
plt.ylabel("Accuracy")
plt.show()
```

```
KNN RESULTS
Accuracy: 0.8522167487684729
Confusion Matrix:
 [[  5   0   0   3   0]
 [  0 110   9   0   0]
 [  0   9  51   0   0]
 [  6   2   0   7   0]
 [  0   0   1   0   0]]
Classification Report:
               precision    recall  f1-score   support

           0       0.45      0.62      0.53         8
           1       0.91      0.92      0.92       119
           2       0.84      0.85      0.84        60
           3       0.70      0.47      0.56        15
           4       0.00      0.00      0.00         1

    accuracy                           0.85       203
   macro avg       0.58      0.57      0.57       203
weighted avg       0.85      0.85      0.85       203


LOGISTIC REGRESSION RESULTS
Accuracy: 0.9605911330049262
Confusion Matrix:
 [[  5   1   0   2   0]
 [  0 118   1   0   0]
 [  0   1  59   0   0]
 [  0   2   0  13   0]
 [  0   0   1   0   0]]
Classification Report:
               precision    recall  f1-score   support

           0       1.00      0.62      0.77         8
           1       0.97      0.99      0.98       119
           2       0.97      0.98      0.98        60
           3       0.87      0.87      0.87        15
           4       0.00      0.00      0.00         1

    accuracy                           0.96       203
   macro avg       0.76      0.69      0.72       203
weighted avg       0.96      0.96      0.96       203
```

```
C:\Users\asus\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\metrics\_classification.py:1833: UndefinedMetricWarning: Precision is ill-def
ined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
C:\Users\asus\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\metrics\_classification.py:1833: UndefinedMetricWarning: Precision is ill-def
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  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
```
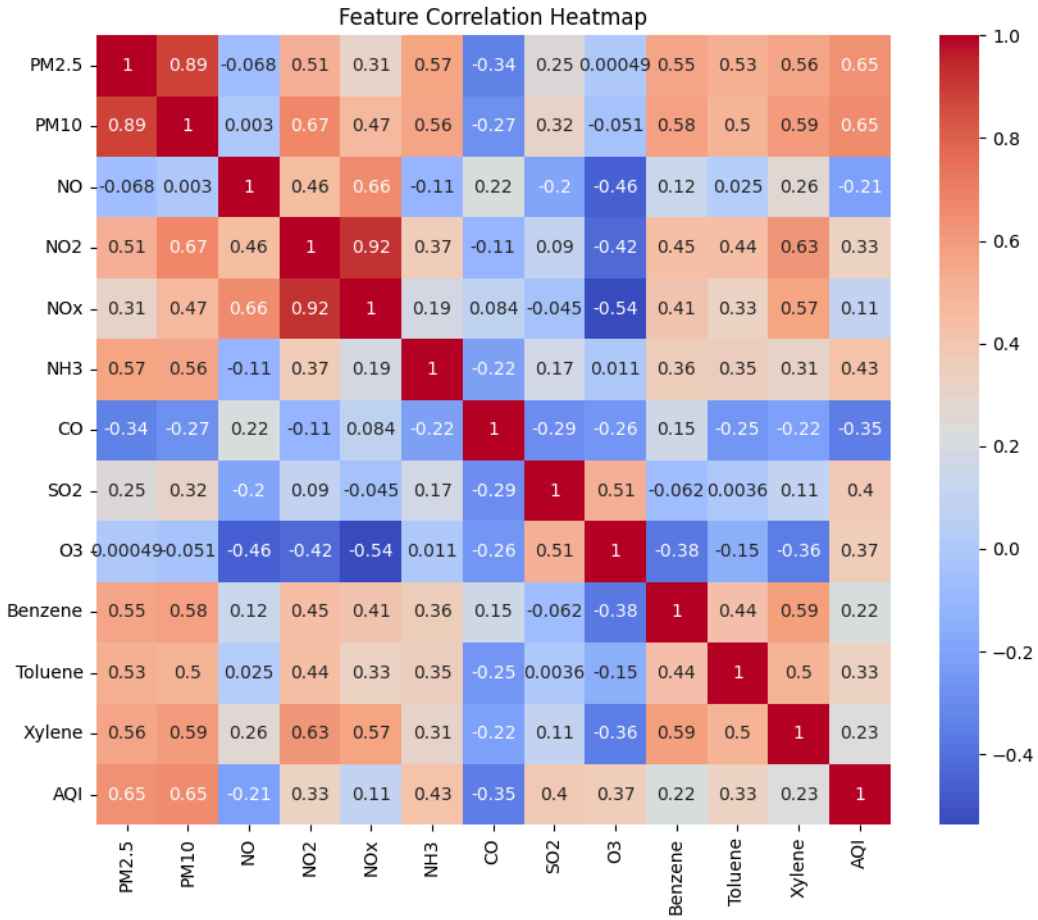
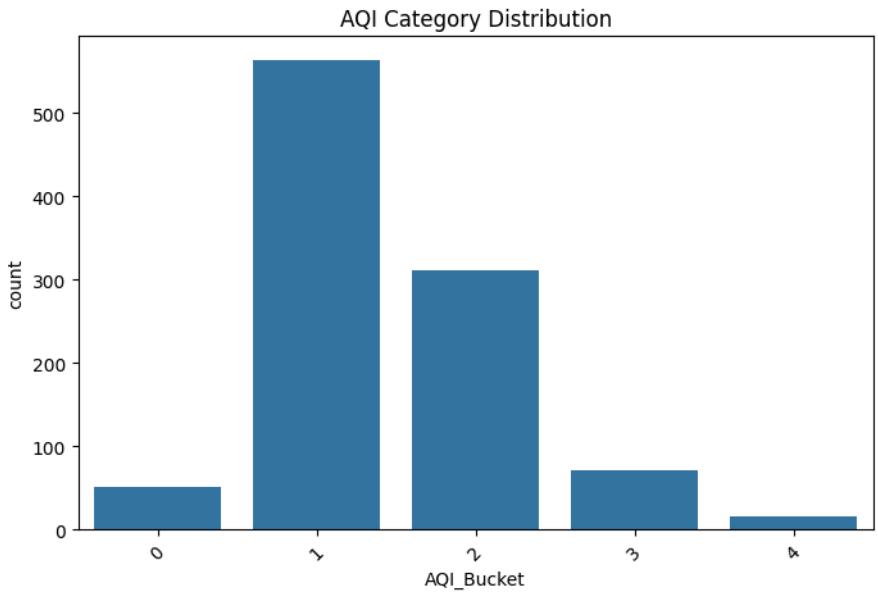
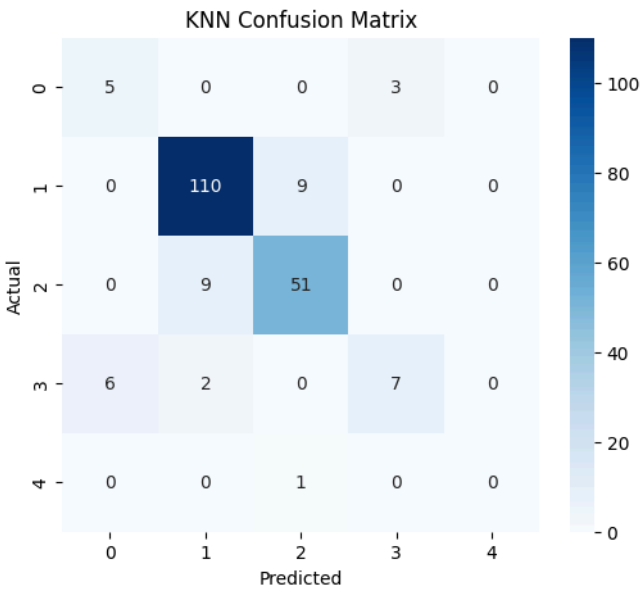
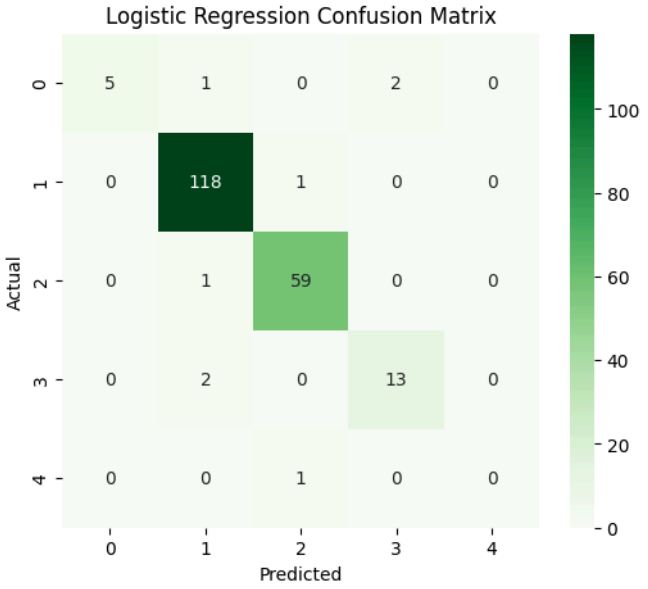
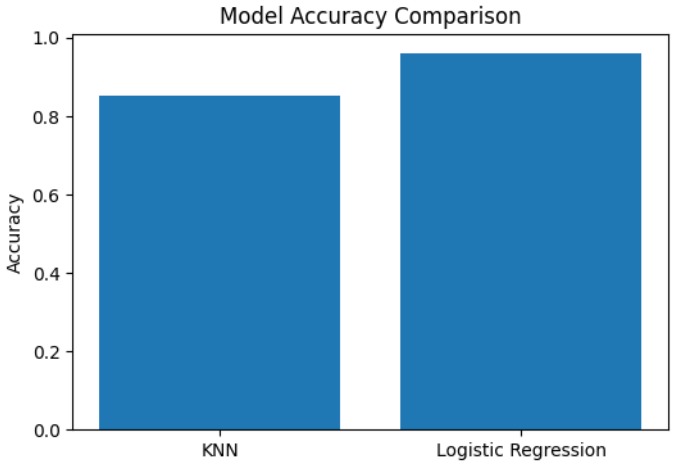


Feature Correlation Heatmap
PM2.5 1 0.89 -0.068 0.51 0.31 0.57 -0.34 0.25 0.00049 0.55 0.53 0.56 0.65
PM10 0.89 1 0.003 0.67 0.47 0.56 -0.27 0.32 -0.051 0.58 0.5 0.59 0.65
NO -0.068 0.003 1 0.46 0.66 -0.11 0.22 -0.2 -0.46 0.12 0.025 0.26 -0.21
NO2 0.51 0.67 0.46 1 0.92 0.37 -0.11 0.09 -0.42 0.45 0.44 0.63 0.33
NOx 0.31 0.47 0.66 0.92 1 0.19 0.084 -0.045 -0.54 0.41 0.33 0.57 0.11
NH3 0.57 0.56 -0.11 0.37 0.19 1 -0.22 0.17 0.011 0.36 0.35 0.31 0.43
CO -0.34 -0.27 0.22 -0.11 0.084 -0.22 1 -0.29 -0.26 0.15 -0.25 -0.22 -0.35
SO2 0.25 0.32 -0.2 0.09 -0.045 0.17 -0.29 1 0.51 -0.062 0.0036 0.11 0.4
O3 0.00049 -0.051 -0.46 -0.42 -0.54 0.011 -0.26 0.51 1 -0.38 -0.15 -0.36 0.37
Benzene 0.55 0.58 0.12 0.45 0.41 0.36 0.15 -0.062 -0.38 1 0.44 0.59 0.22
Toluene 0.53 0.5 0.025 0.44 0.33 0.35 -0.25 0.0036 -0.15 0.44 1 0.5 0.33
Xylene 0.56 0.59 0.26 0.63 0.57 0.31 -0.22 0.11 -0.36 0.59 0.5 1 0.23
AQI 0.65 0.65 -0.21 0.33 0.11 0.43 -0.35 0.4 0.37 0.22 0.33 0.23 1
PM2.5 PM10 NO NO2 NOx NH3 CO SO2 O3 Benzene Toluene Xylene AQI
1.0 0.8 0.6 0.4 0.2 0.0 −0.2 −0.4


AQI Category Distribution
count
0 100 200 300 400 500
0 1 2 3 4
AQI_Bucket


KNN Confusion Matrix
Actual
0 5 0 0 3 0
1 0 110 9 0 0
2 0 9 51 0 0
3 6 2 0 7 0
4 0 0 1 0 0
0 1 2 3 4
Predicted
0 20 40 60 80 100


Logistic Regression Confusion Matrix
Actual
Predicted
0
1
2
3
4
5 1 0 2 0
0 118 1 0 0
0 1 59 0 0
0 2 0 13 0
0 0 1 0 0
100
80
60
40
20
0


Model Accuracy Comparison
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
KNN
Logistic Regression

In [ ]: